

Building Critical Thinking with AI Output Validation Tasks

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Outline



- Icebreaker
- Background
- Group Work: Output Validation Task
- Assessing Critical Thinking: Preliminary Findings
- Application to Curriculum Design

By the end of the workshop, you will....

 Consider the relationship between humans and AI tools

 Engage in a classroom task to encourage learner agency

 Be excited to teach critical thinking using AI tools!



You are welcome to take photos during today's presentation.

I will also share the slides and materials at the end.

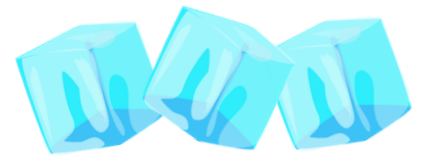


You are welcome to re-use and share these slides and activities.



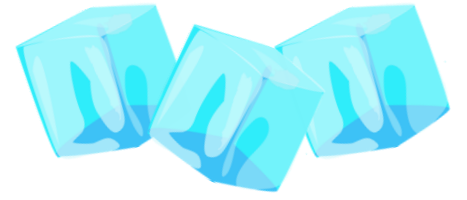
Icebreaker: Didn't Live Up to the Hype

#17 Ordered A Beautiful Chrome Doberman Statue, Description Said 3ft Tall





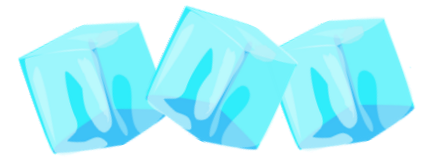




In Groups

- Say “hello” to everyone in your group.
- Choose one person to be the moderator.
 - You will keep the group on task.

In Groups



- Think of a time you bought something online and what arrived was nothing like what you expected.
- Use these starters: *"I ordered... but what I got was..."* or *"I thought it would be... but actually..."*
- Everyone shares one story with the group.
- **Moderator**, be sure everyone shares something.
- *No shopping stories?* Think of a time something was not as good as you expected.

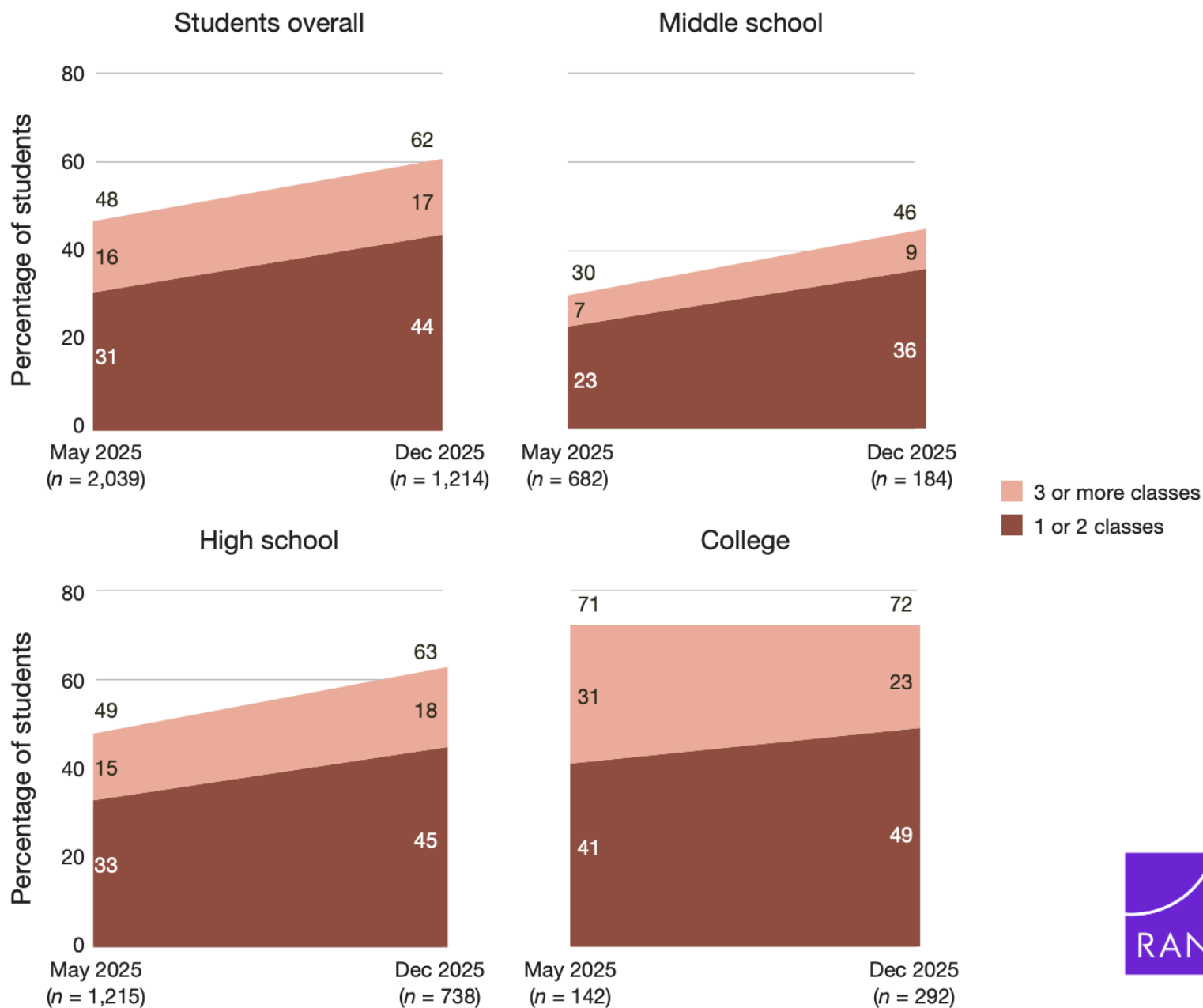
All of these “shopping fails” are examples of a lack of **quality control**.

This is why we need **humans in the loop**.



Background

FIGURE 1
Students Increasingly Use AI Help for Homework



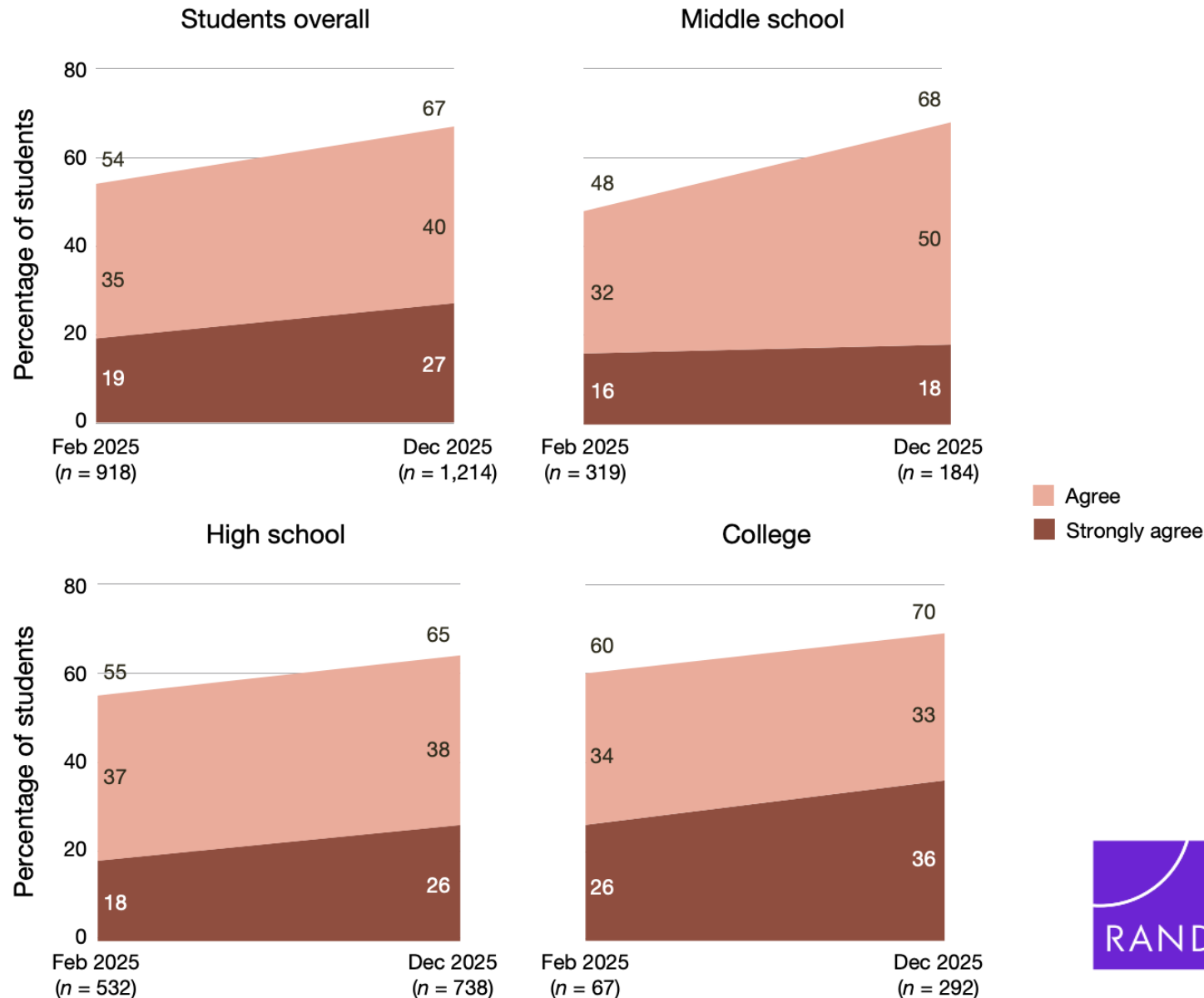
AI use for homework is rising across all grade levels.



(Schwartz & Diliberti 2026)

FIGURE 2

Students Increasingly Agreed That Using AI for Schoolwork Harms Critical Thinking Skills



**More students
believe AI
harms critical
thinking.**

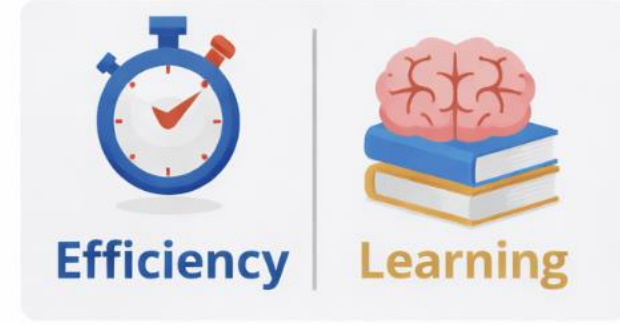


(Schwartz & Diliberti 2026)

Why bring a forklift to the gym?



Framing the Problem



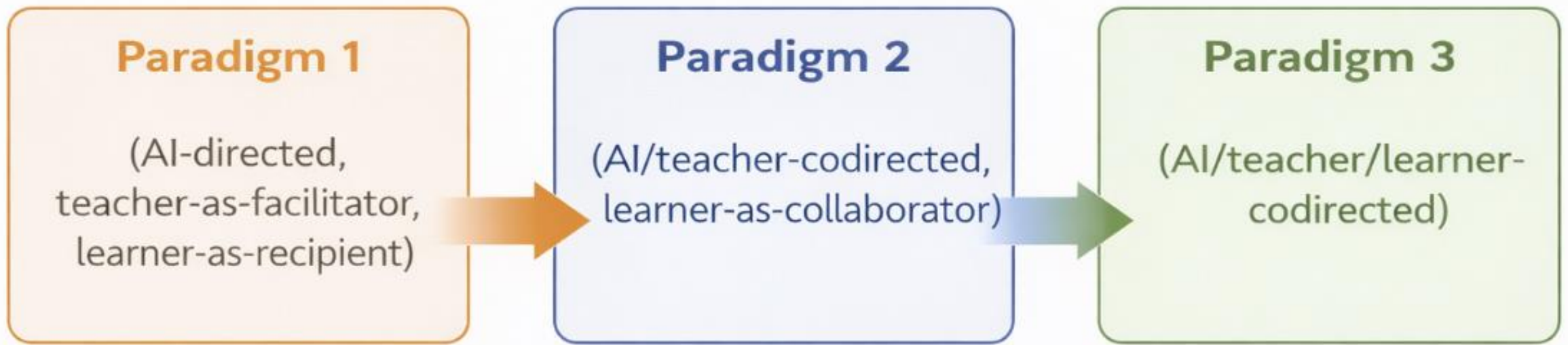
Much of the literature has focused on how AI makes learning more **efficient** for students, and more research is needed in how AI tools can be applied to **help students learn** (Crompton & Burke 2025).

Human-AI Interaction (HAI)



- Language learning as meaning making is shifting from solely human-to-human interactions to a broader understanding of human-to-computer interactions (Jiang & Hyland, 2026)
- AI-, teacher-, and learner-directed task structures support different forms of language learning and critical thinking (Wang et al., 2024)

Human-AI Interaction (HAI)



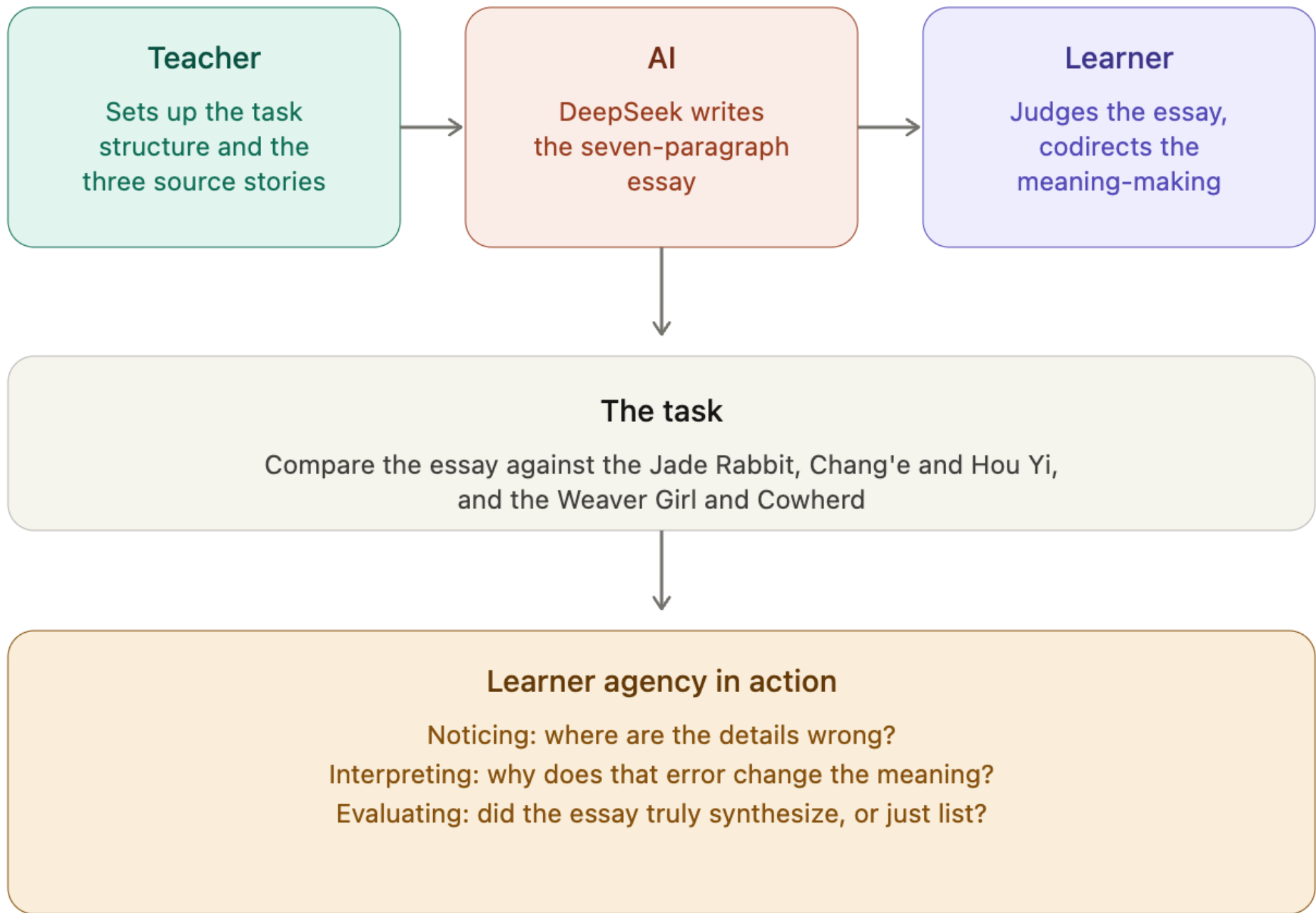
Students can alternate between paradigms depending on the task.

Paradigm 3 has the potential for the most advanced and critical stage of AI use for students.

(Wang et al., 2024)

Group Work:

Output Validation Task

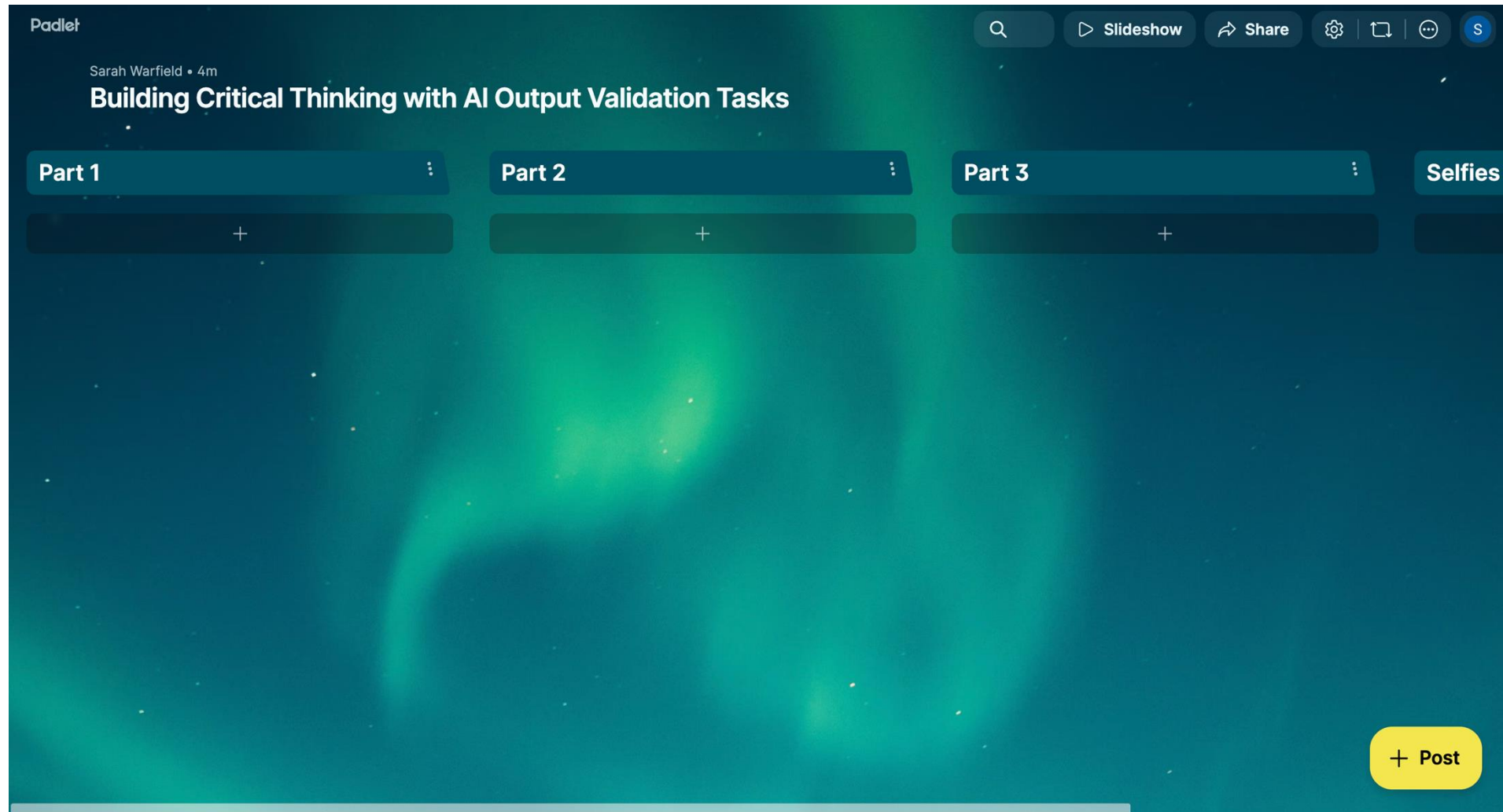


In Pairs

- Skim the Deepseek versions of the three folk tales
- Skim the Deepseek-generated essay
- Guiding Questions
 - *How well do you know each story?*
 - *What do you remember from learning the story before?*
 - *How well-written is the Deepseek-generated essay?*



Scan the Padlet



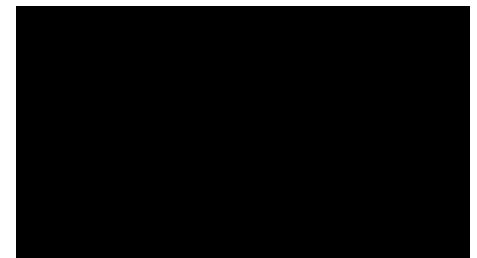
Create a new post to write down your answers—choose one recorder
Feel free to comment on the posts of others
Extra column for Selfies

Part 1: Finding mistakes in details



What mistakes did the essay make in reporting facts or details?

- ☰ List the mistakes you find
- ✓ Say what the correct answer should be

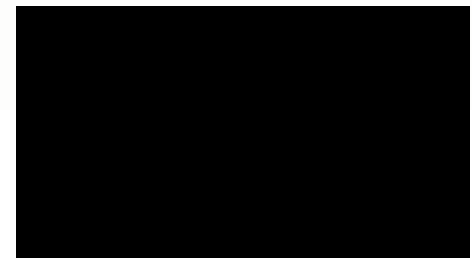


Part 2: Finding synthesis mistakes

 Find two examples where the essay did not correctly synthesize across the three stories.

-  What was the mistake?
-  Where in the original stories is the evidence?
-  What should the essay have said instead?

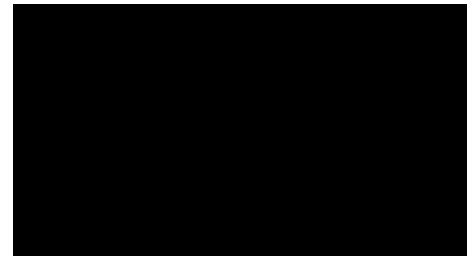
✓ No mistakes? List two examples where the essay did synthesize well instead.



Part 3: Open-ended Questions

 Take a minute to think about how the essay organized and connected the three stories. What did you learn or notice that you hadn't thought of before?

99 Did you notice any phrases the essay used to transition between stories or tie ideas together? What did you think of them?



Learner Agency in Paradigm 3

- Turn and Talk
- *How did you engage with the experience of “learner agency” in this task?*
- *Did it feel like a critical thinking task? Why or why not?*
- *How would you assess critical thinking?*

Expansions

- Jigsaw activity in class
 - Each group/pair has an AI-generated essay to check for mistakes (across different sources)
 - Share out results in groups
- Create just the essay, not the video, and do the same task
- Focus on just one aspect (meaning, details, facts, synthesis)
- Check accuracy against student-sourced materials

Assessing Critical Thinking: Preliminary Findings

Unit 1 Review: Output Validation

Everyone who completes this form by Monday at 10PM will get one extra point on the midterm listening exam.

Directions:

Watch the NotebookLM Video for Unit 1 [here](#). While you watch, notice (a) any information that the video correctly states and (b) any information that the video incorrectly states. When you are noticing this information, it can be facts, citations, or points of synthesis. Regarding the latter category, points of synthesis, notice how well—or how poorly—the AI tool can synthesize across sources.

Keep good notes on these two points as you will refer to them later. I will also ask you to upload a copy of your notes.

Tip: You can also listen for any useful phrases for signposting or organizing information that NotebookLM uses. I'll ask you about those too.

Remember: When you are looking for inconsistencies it can be either the AI tool is *wrong* in how it understands the source material or it is *wrong* in how it analyzes across sources. The analysis of sources is a personal, critical thinking skill. Perhaps the AI tool synthesizes sources in a way that you do not find correct. If that is the case, be prepared to tell me why.

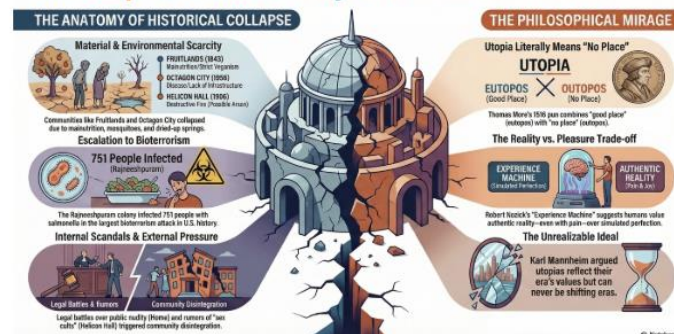
Follow the instructions for each question and then type your answer. Click "submit" when you are finished. **You will need to access all the materials from Unit 1 to answer the questions.**

* Indicates required question

Email *

Your email

The Utopian Paradox: Why Perfection Fails in Practice



- Materials review before midterm listening test
- Watch a NotebookLM-generated synthesis video of the course materials and check it for accuracy



The screenshot shows the NotebookLM interface with the following components:

- Sources:** A list of sources used for the synthesis, including 'The Resurgence of Marx in...'.
- Chat:** A chat window titled 'Unit 4 Review' showing a synthesis of the provided text. The text discusses the resurgence of Marxist ideology, its relevance to contemporary issues like wealth inequality and labor instability, and the challenges of applying these theories in the modern world.
- Studio > Video Overview:** A video player showing a video titled 'The Resurgence of Karl Marx' based on 1 source. The video features a portrait of Karl Marx and a timeline of his life.



AI Output



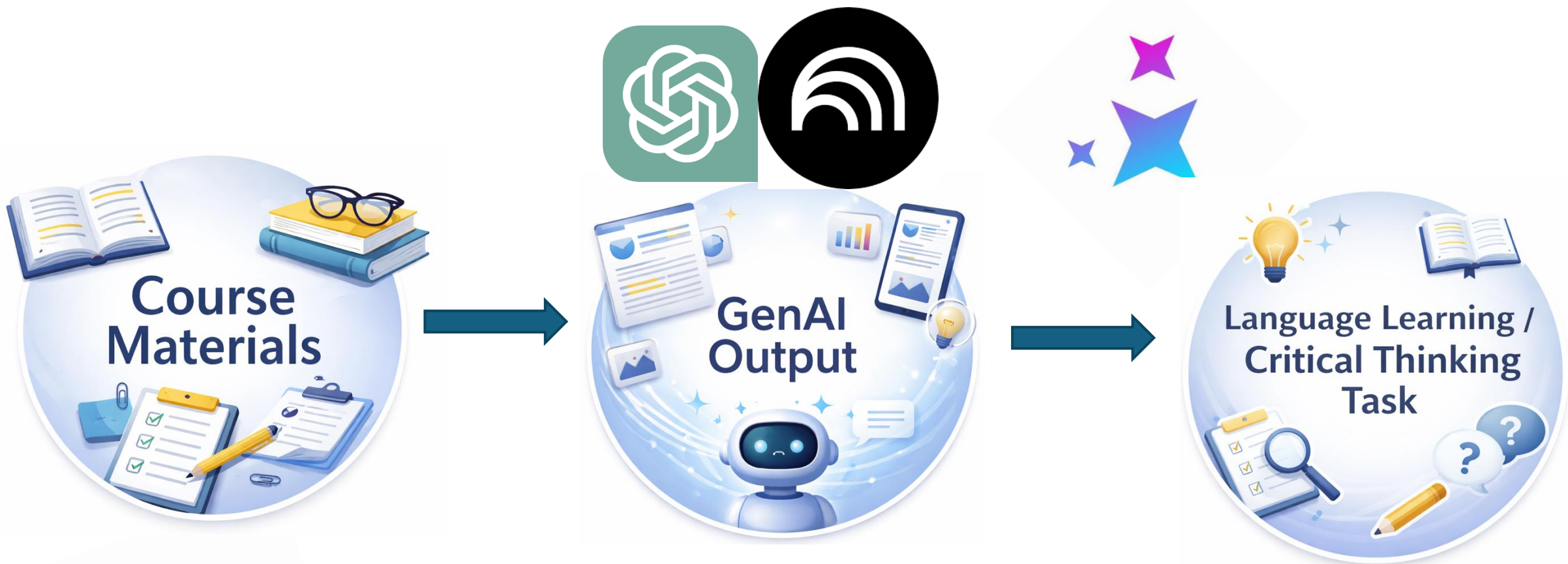
Original Materials



**Critical
Thinking**

Theoretical Rationale

- Students negotiate meaning through well-structured tasks (Ellis et al., 2019)
- By engaging with AI output in a structured way, output becomes a process rather than just a product (Swain, 2005)



Creating the Output Validation Task



Resources ▾ Enterprise Pricing

h Zapier ▾ Productivity ▾ Business growth ▾ Product & platform ▾ Company updates ▾

Claude vs. ChatGPT at a glance

Claude and ChatGPT are powered by similarly capable [LLMs](#) and [LMMs](#). They differ in some important ways, though:

- ChatGPT is best for **users who want an all-in-one AI toolkit**. Its image and video generation capabilities, custom chatbots, and browser-based agentic automation make it ideal for users who want to explore the full spectrum of what AI can do.
- Claude is best for **users focused on coding and sophisticated creative work**. Its more natural writing style, powerful agentic coding features, and thoughtful analytical approach make it the best choice for developers, writers, and analysts who need depth over breadth.



**Use a trusted LLM when
creating tasks for students.**

AI, Power, & Our Cities



The Evolution of AI

Traditional

Humans write explicit instructions for the computer to follow.

Deep Learning

AI learns by finding patterns in vast amounts of data on its own.

Reinforcement

AI learns from feedback and rewires its own thinking, like a child.



Local councils and staff will **always** be **accountable** for the AI systems they use.



What mistakes did NotebookLM make in reporting facts or citations? List them below and then tell me what the correct answer is.

Description (optional)

Mistake One

Short answer text

Mistake Two

Short answer text



Student Results (n=7)

Unit 1

- 4 Meaning
- 1 Citation
- 7 Details

Unit 2

- 6 Meaning
- 1 Citation
- 9 Details

Find two examples where NotebookLM did **not** correctly synthesize across sources.



For each example, tell me (1) what the mistake was, and then tell me (2) *where* in the original source you found evidence that NotebookLM made a mistake, and finally (3) tell me what the audio overview should have said.

If you think there are no mistakes, you can instead list two examples where NotebookLM did not make mistakes synthesizing across sources.

Example 1

Long answer text

Example 2

Long answer text



Student Results (n=7)

- Unit 1: 14 responses
 - 0 noticing of a synthesis mistake (0% of responses)
- Unit 2: 11 responses
 - 8 noticing of a synthesis mistake (73% of responses)

This is the final section.



Description (optional)

Take a minute to think about how NotebookLM organized and delivered the information. Tell me, in your own words, learned about the Unit 1 materials that you did not know before listening to this video.

Long answer text

Did you hear any phrases for signposting or organization that you would like to use? Tell me about them in the space below.

Long answer text

Preliminary Results

- Finding mistakes in **details** allows students the chance to notice key language, concepts, and how ideas are actually expressed in the original texts, supporting language learning (Schmidt, 1990)
- The evidence of students explaining **synthesis** mistakes suggests they are developing academic literacy, particularly in interpreting and evaluating meaning across texts (Hyland, 2004)

Preliminary Results



- Ngo & Hastie (2025) found that well-designed pedagogic interventions led to **critical thinking gains** among EAP students who were tasked with analyzing GenAI results in a learning task.
- Tasks that are designed to force students to use metalinguage —> instructors assess deeper learning (Mishra et al., 2023)

Application to Curriculum Design

Teaching Metalanguage Directly



Introduce Terms

Give students words for noticing, interpreting, evaluating



Practice the Language

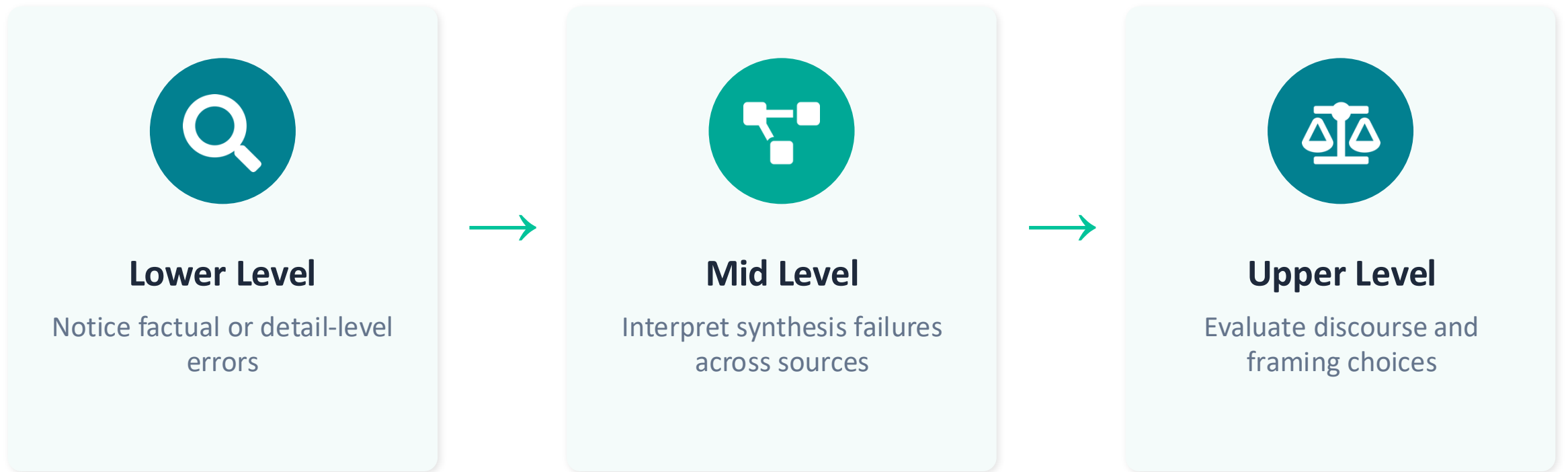
Use those terms inside the validation task itself



Talk About Learning

Apply the language to their own learning, not just the AI output

Sequencing Across a Program



One activity becomes a developmental thread across an entire program.

Building Faculty Capacity, Not Just One Task



One Instructor

Designs a task that works in their own classroom



Shared Standards

Calibrate what counts as good noticing vs. interpreting



Whole Department

Other instructors design their own versions of the task

The unit of intervention shifts from a classroom activity to a faculty development effort.

Making Critical Thinking Visible at the Program Level



Student Output

Explanations of AI errors
generate written evidence



Scored Evidence

Evidence is scored against a
critical thinking outcome



Program Assessment

Feeds into the assessment
cycle the program already
runs

By the end of the workshop, you will....

 Consider the relationship between humans and AI tools

 Engage in a classroom task to encourage learner agency

 Be excited to teach critical thinking using AI tools!

Thank you!

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Today's materials



sarah

Richmond, United Sta...



Scan QR code to add me as a friend.

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